UPI FRAUD DETECTION USING MACHINE LEARNING

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***Abstract*— Unified Payments Interface (UPI) operations have emerged as a major target for fraudulent activity due to the quick adoption of electronic payment methods presenting serious security and financial threats to both users and banks. Significant financial losses result from traditional rule-based identification of fraud system's frequent inability to adjust to changing fraudulent trends. This paper suggests a cutting-edge machine learning-powered UPI fraud detection technique that uses strong classification models to successfully detect and stop fraudulent transactions in order to address this urgent problem. To find patterns and connections a fraud dataset from Kaggle is used which is carefully preprocessed before undergoing in-depth Exploratory Data Analysis (EDA). The robustness of the model is ensured by using outlier detection techniques to find anomalies in transaction data. To maximize model generalization the data into training and testing halves in an 80-20 ratio. Decision Tree, Random Forest, Gradient Boosting and AdaBoost are among the machine learning models that are used, and they all achieve remarkable accuracy scores of 99.9690%, 99.7%, 99.9% and 99.75%, respectively. With the help of Streamlit, models that have been trained are stored and easily incorporated into a real-time detection of fraud system offering an intuitive user interface that facilitates immediate fraud detection. In addition to improving security this approach gives financial institutions a clever, scalable and flexible way to successfully fight UPI fraud which promotes dependability and trust in digital payment networks.**

Keywords*— *Anomaly Detection, Digital Payments, Fraud Detection, Machine Learning, Transaction Security and UPI Fraud.

I. INTRODUCTION

Financial transactions have been completely transformed by the development of digital payment systems which provide consumers with an easy and quick way to send money [1]. Because of its ease of use, real-time processing and compatibility with various financial platforms the Unified Payments Interface (UPI) which was initially launched by the National Payments Corporation of India (NPCI) in 2016 has grown to be one of the most popular digital payment methods. UPI has become a popular method of financial exchange as a result of its exponential growth in transactions over time. But along with this quick expansion fraudsters have also taken advantage of system flaws which has resulted in a startling increase in UPI frauds [2]. Fraudsters utilize a variety of dishonest tactics such as virus attacks, phishing, identity theft and illegal access to account information to cause large financial losses and undermine confidence in digital payment systems.

The majority of current fraud detection systems are built on heuristic and rule-based algorithms which are not flexible enough to accommodate new fraudulent strategies. These conventional techniques frequently generate a large percentage of errors or are unable to identify complex fraud patterns which results in ineffective fraud protection [3]. Furthermore traditional security measures are less successful against fraud schemes that are constantly changing since they rely on static thresholds and established rules. Although several financial organizations have used simple machine learning models these models frequently need detailed feature engineering and do not fully utilize sophisticated classification algorithms. To ensure secure transactions it is now essential to have an intelligent, computerized and real-time fraudulent detection system that can learn from data about transactions dynamically and accurately identify fraudulent activity. To enhance data quality and spot questionable transaction patterns the methodology [4] uses data pretreatment, exploratory data analysis (EDA) and outlier detection.

By enhancing financial security and protecting users from fraud related losses our effort offers significant advantages that go beyond technological improvements. Ensuring transactional safety is essential for financial stability and digital electronic transactions become the foundation of contemporary economies. Millions of users particularly those who are not aware with cybersecurity procedures will be shielded from online fraud by a strong fraud detection system. Furthermore by incorporating this technology into their current frameworks for preventing fraud, financial companies and normal agencies can improve the workload associated with manual investigations into fraud and increase the effectiveness of fraud detection efforts. This research helps create a more secure, robust and fraud-free financial system by promoting trust in electronic payment platforms. The growing number of UPI fraud cases [5] which jeopardize the security of electronic transactions and erode user trust in

online payment systems is what spurred this study. Due to a lack of an effective fraud detection system many people and businesses still experience financial losses and unlawful transactions despite cybersecurity developments. An automated, intelligent and flexible prevention system that can identify and stop fraud in real time is required due to the increasing complexity of fraudulent schemes [6]. This study intends to close the gap in current fraud detection techniques by leveraging machine learning along with artificial intelligence providing users and businesses with a dependable and expandable solution.

II. LITERATURE SURVEY

Using conventional rule-based and statistical techniques fraud detection in electronic payment systems has been the subject of several previous research. In order to detect suspicious activity early fraud detection techniques used manually established rules including transaction value limitations, frequency constraints, and recognized fraud patterns. Although somewhat successful these systems had trouble adapting as scammers kept changing their strategies. In order to detect fraudulent transactions based on departures from typical spending patterns researchers then used mathematical models such as logistic regression and anomaly detection approaches. Nevertheless, these models frequently had large false-positive rates and were unable to generalize to other types of fraud. Jagadeesan et al. although the UPI has transformed digital transactions in India it also offers the possibility of fraudulent activity. By employing unsupervised, supervised and semi-supervised methods to analyze transaction data machine learning techniques can assist in detecting UPI fraud [7]. While continuous tracking and adaptive learning strategies enhance security and user trust, effective choice of features and engineering techniques improve performance. Gupta et al. combines machine learning and rule-based tactics is being developed [8]. In order to improve user trust and make UPI transactions more secure, this system employs anomaly detection algorithms to identify questionable transactions and user behavior.

Rani et al. incorporated into a real-time UPI payment tracking device recognizing unusual activity with 98.2% accuracy [9]. This increases UPI safety during transactions and the modern financial era by showing the effectiveness of algorithmic learning in identifying fraudulent transactions. Although supervised models that were trained on labelled fraud datasets have demonstrated increased accuracy they still need to be updated often with fresh fraud cases. Novel fraud patterns have been found using unsupervised methods like autoencoders and clustering that don't require prior labeling [10]. Even said there are still issues with current systems including unbalanced datasets, excessive processing costs and the incapacity to identify complex fraud efforts instantly. Many fraud detection systems are useless for preventing fraud right away because they lack strong real-time processing capabilities. Furthermore issues with interpretability, scalability and legal compliance continue to hinder the integration of AI-based detection of fraud into financial systems.

III. DATA COLLECTION & PREPROCESSING

The dataset which comes from Kaggle includes transactional information pertaining to UPI payments along with labels designating whether the transaction is authentic or fraudulent. Transaction ID, timestamp, hidden sender and recipient information, transaction amount, transaction kind and status are just a few of the features included in this dataset. Furthermore metadata like IP addresses, device IDs and geolocation information aid in the detection of fraudulent activity. Preprocessing is a crucial step to guarantee that high-quality data is supplied into the models for statistical learning because real time financial data mostly contains noise, missing values and errors. Imputation approaches such as mean/mode imputation for numerical features and forward-fill or backward-fill procedures for time-series data are first used to address missing values. To avoid bias in model training, duplicate transactions which may occur as a result of system errors are also eliminated.

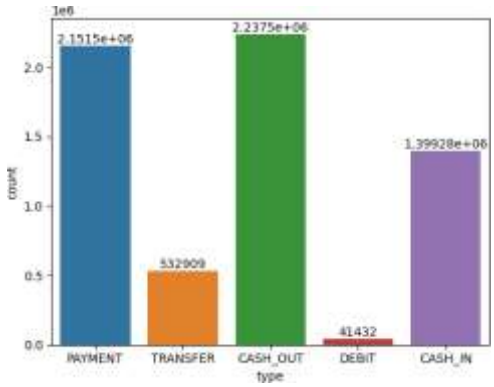

Fig.1 Count of Types of Transactions

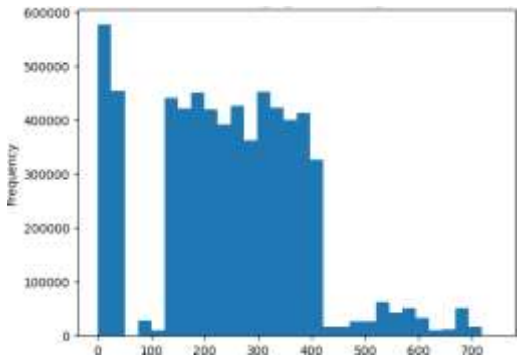

Fig.2 Distribution of Transaction Steps

To standardize the data and enhance model performance, normalization and scale of features are carried out. Min-max scaling is used to ensure that values stay within a predetermined range (0 to 1) because transaction amounts can vary greatly. To ensure that features with different distributions contribute similarly during model training standardization approaches like Z-score normalization are employed. Since fraudulent transactions frequently show unusual patterns in comparison to legitimate transactions identifying outliers is another essential step. New useful features such average transaction amount, transaction frequency per user and velocity characteristics that gauge abrupt increases in transaction volume are also created through feature engineering [11].

Several data processing methods are used to organize the dataset for various modeling and analytical uses in order to provide the best possible fraud detection. To investigate feature correlations and identify abnormalities, visualization tools including correlation heatmaps, box plots and histograms are employed. The most pertinent characteristics are kept while redundant or less instructive ones are removed using feature selection approaches like mutual information gain and recursive feature elimination (RFE). Unbalanced datasets a prevalent problem in identifying fraud where the number of phony transactions is much smaller than the number of valid transactions are addressed using class balancing strategies [12].

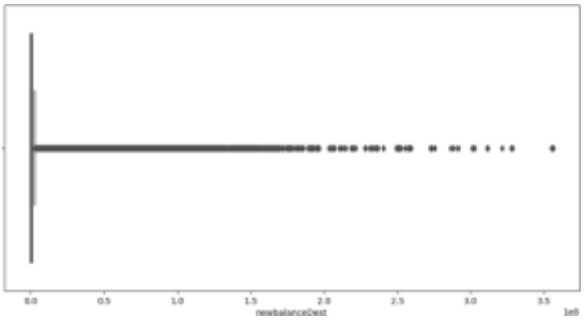

Fig.3 Outlier Detection in Transactions

To properly assess model performance the data is separated into subsets for training and testing following feature engineering and data preprocessing. The machine learning models are trained using 80:20 train-test split with the remaining 20 percent set aside for testing and validation. Stratified sampling is utilized to maintain the ratio of fraudulent and genuine activities in both training and testing sets, preventing data leakage and guaranteeing that the model performs effectively when applied to testing data. To adjust the model parameters and avoid overfitting, cross-validation methods like k-fold cross-validation (with k=5 or 10) are used. To maximize model performance, search of grids and random search are used for hyperparameter tuning [13].

IV. PROPOSED METHODOLOGY

Before moving on to model training and deployment the suggested technique for UPI fraud detection follows a systematic pipeline that starts with data gathering, preprocessing and exploratory data analysis (EDA). The Kaggle dataset is subjected to extensive preparation, which includes normalizing numerical characteristics, managing missing values and encoding categorical variables. To eliminate unusual data points that could distort the model outlier identification methods like Isolation Forest and Interquartile Range (IQR) are used. Furthermore feature engineering is used to generate new variables that improve the model's capacity to identify fraudulent transactions such as velocity-based features, average transactions amount and transaction frequency per user.

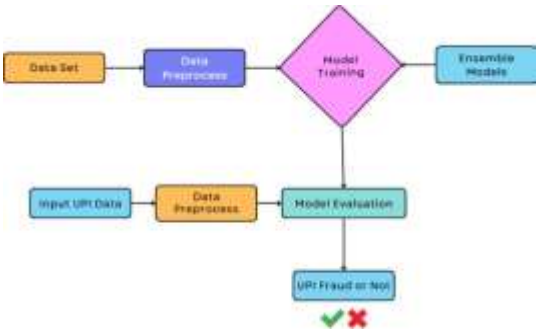

Fig.4 Working Methodology

Machine learning models including as Decision Tree, Random Forest, Gradient Boosting and AdaBoost are used to categorize transactions as either valid or fraudulent after the data has been prepared. Grid Search and Random Search are used for hyperparameter tuning to maximize model performance while k-fold cross-validation is used to guarantee robustness and avoid overfitting. The models are assessed using performance criteria like precision, recall, precision, F1-score and AUC-ROC the top-performing models had a precision rate of 99.9690%, 99.7%, 99.9% and 99.75% respectively. Following training the models are saved and incorporated into Streamlit a real-time fraud detection system that offers an interactive interface for users to enter transaction information and get instant fraud risk estimates. The security of online transactions is improved by this end-to-end method which guarantees a scalable, precise and effective fraud detection framework.

A. Decision Tree (DT)

With regard to its ease of use and capacity for handling both numerical and category-based information decision trees a basic supervised learning technique used for regression and classification tasks are very useful in fraud detection [14]. The decision-making process is simple to understand and describe since it is based on a sequence of "if-else" situations. Because decision trees can find patterns that differentiate fraudulent transactions from genuine ones and capture complicated linkages in transactional data they are very helpful for fraud detection. The capacity of decision trees to successfully manage outliers and missing variables is one of their main advantages. Decision Trees are free of feature scaling or normalization in contrast to linear models that presume a particular data distribution. Without requiring a lot of preparation they can function effectively with both big and small datasets. Decision trees also provide feature priority ranking, which helps fraud investigators pinpoint the most important characteristics that lead to fraudulent transactions like high transaction frequencies strange transaction amounts or irregular geographical locations. Decision trees do have certain drawbacks though especially their propensity to overfit when working with big and complicated datasets.

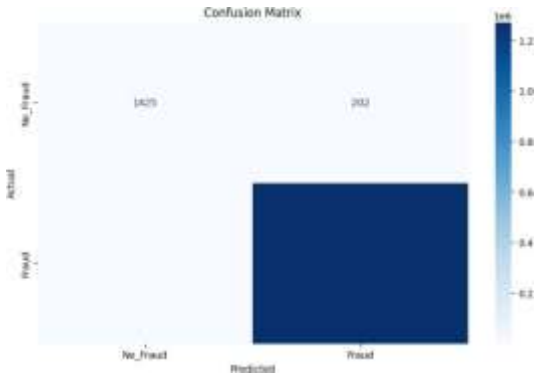

Fig.4 Confusion Matrix of Decision Tree Classifier

When a decision tree gets overfit it performs poorly on unseen data because it learns from noise instead of generalizable patterns. Techniques like pruning the removal of superfluous branches and imposing restrictions on tree height are sometimes employed to avoid overfitting. The sensitivity of decision trees to minute changes in data is another difficulty. Model stability can be diminished by a small alteration to the dataset which can result in an entirely new tree structure. These problems are addressed by ensemble techniques like Random Forest and Gradient Boosting, which combine several Decision Trees to enhance generalization. The Decision Tree framework is still a powerful foundational method for fraud detection in spite of its drawbacks. It provides the framework for more sophisticated ensemble methods that build on its ideas while addressing its drawbacks like Random Forests and Gradient Boosting. Decision trees are very helpful for rule-based fraudulent classification in UPI fraud detection because they offer a clear decision-making process that analysts in finance and fraud detection teams can readily understand.

B. Random Forest (RF)

By merging several decision trees Random Forest is a collective learning approach that enhances the Decision Tree model's accuracy and resilience [15]. Random Forest creates a "forest" of trees by randomly choosing subsets of characteristics and training each tree separately in contrast to a single Decision Tree which may experience overfitting. Through majority voting (for classification) or average (for regression), the predictions of each individual tree are combined to get the final forecast. This method is quite useful for detecting fraud in UPI transactions since it improves model stability and generalization. The capacity of Random Forest to manage high-dimensional datasets with a large number of characteristics is one of its main advantages. The model lowers feature correlation, avoids overfitting and enhances performance on unknown data because each tree has been taught on a random sample of features. This functionality is especially helpful in fraud detection, as a variety of transactional variables, including device ID, user location, transaction amount and transaction time all contribute to the categorization process. Relative to a single Decision Tree, Random Forest offers a more thorough analysis by taking into account several factors of a transaction.

Random Forest's resilience to noise and values that are missing is an additional benefit. The ensemble aspect of Random Forest guarantees that small inconsistencies do not substantially affect overall performance in contrast to individual Decision Trees which can be quite sensitive to slight differences in the data. Additionally the model offers feature importance rankings which enable fraud analysts to pinpoint the main causes of fraud. In banking applications where understanding is essential for decision-making and regulatory compliance, this transparency is vital. Despite its benefits, Random Forest has certain drawbacks such as longer prediction times when compared with simpler models and more computational complexity. Real-time fraud detection is more resource-intensive since the model needs greater processing and memory capacity due to the construction of numerous trees. Furthermore even if Random Forest increases accuracy it might not be able to handle extremely unbalanced datasets where forged payments are uncommon. To increase its efficacy in fraud detection settings strategies including threshold tuning, SMOTE and class weighting are frequently employed.

C. AdaBoost (AB)

Another ensemble learning technique called AdaBoost (Adaptive Boosting) combines several weak learners typically shallow Decision Trees to create a powerful prediction model increasing classification accuracy. AdaBoost gives each training instance a variable weight, making misclassified data more significant in each iteration than Random Forest and Gradient Boosting which concentrate on aggregating numerous trees. The model is quite successful at detecting fraud because of its adaptive methodology which enables it to concentrate on cases that are challenging to categorize. AdaBoost's capacity to lessen bias while preserving interpretability is one of its main benefits. Because the model is based on weak classifiers (usually single-level Decision Trees called "stumps") it is straightforward but efficient. Because of this feature AdaBoost is especially helpful in fraud detection, where simple decision rules can assist in accurately identifying fraudulent transactions. AdaBoost makes sure that mistakes from earlier iterations are fixed by iteratively improving the model which improves classification performance.

Additionally AdaBoost is resistant to overfitting particularly in contrast to deep decision tree structures. It retains computational efficiency while generalizing effectively to new data since it constructs several shallow trees as opposed to deep ones. AdaBoost is a versatile option for fraud detection applications since it can also be used with base learners other than decision trees such as support vector algorithms (SVMs) and neural networks. It can adjust to diverse fraud detection settings and datasets by incorporating different classifiers. AdaBoost does have certain drawbacks though such as its sensitivity to outliers and noisy data. Extreme outliers have the potential to substantially affect the model resulting in decreased accuracy because the algorithm gives misclassified occurrences more weights. Furthermore unless classes weighting or resampling methods are used AdaBoost may have trouble processing severely imbalanced fraud datasets, even when it performs well on balanced

datasets. AdaBoost's great precision, flexibility and capacity to concentrate on hard-to-detect fraudulent transactions make it a formidable candidate for fraud detection in spite of these obstacles.

D. Gradient Boosting (GB)

Gradient Boosting is very useful for complicated classification tasks where subtle trends need to be found, such fraud detection because of its stepwise development approach. The capacity of gradient boosting to identify complex associations in the data is one of its key benefits. The model learns from errors and improves its predictions after several iterations by concentrating on cases that were incorrectly classified. Because of its adaptability it can identify complex fraud patterns that more basic models would miss. Because of its high sensitivity to minute irregularities gradient boosting is a useful technique in UPI fraud detection since fraudulent transactions frequently resemble normal activity.

Compared to conventional Decision Trees Gradient Boosting is less likely to overfit and performs exceptionally well when working with high-dimensional data. Gradient Boosting avoids overcomplication and guarantees improved generalization by implementing regularization strategies like learning rate regulation and tree depth limitations. Large-scale financial datasets can benefit from the enhanced efficiency and scalability of more sophisticated gradient boosting variants like XGBoost, LightGBM and CatBoost. Gradient Boosting does have certain disadvantages though such as increased training times and a high computational cost. The model necessitates sequential instruction which can be time-consuming because each tree is dependent on the performance of the trees that came before it. Furthermore obtaining optimal performance necessitates rigorous experimentation with hyperparameter tweaking (e.g., learning rate, number of estimation methods and maximum depth). Gradient Boosting's versatility and high accuracy in predicting make it one of the best machine learning models in fraud detection in spite of these difficulties.

V. RESULTS

Several performance criteria such as accuracy, precision, recall, F1-score and AUC-ROC curves were used to assess the fraud detection models. With an exceptional accuracy of 99.9690%, the Decision Tree model followed by Random Forest (99.7%), Gradient Boosting (99.9%) and AdaBoost (99.75%). These high accuracy numbers show how well the models can differentiate between transactions that are fraudulent and those that are legitimate. However as fraud data is frequently extremely unbalanced accuracy alone is insufficient to assess fraud detection methods. Therefore in order to make sure that the models successfully detected fraudulent transactions without producing an excessive number of false positives, precision and recall were also examined.

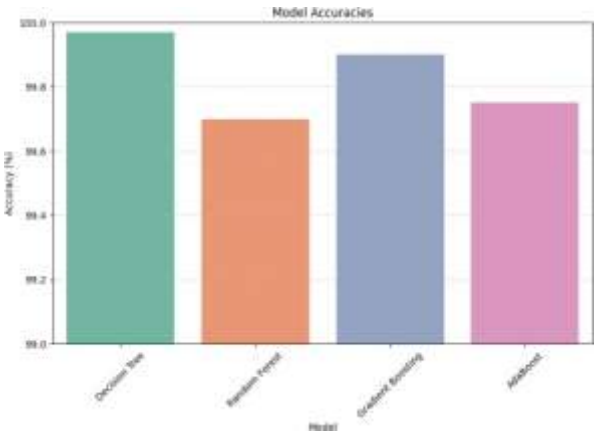

Fig.5 Comparison of Model Accuracies

Plots such as confusion matrices that ROC (Receiver Operating Characteristic) curves and importance of features graphs were created in order to better comprehend the model's performance. The number of actual positives, false positives, genuine negatives and false negatives for each model was revealed by the confusion matrix. As demonstrated by its comparatively greater percentage of false positives, the Decision Tree model exhibited a small propensity toward overfitting despite its high accuracy. However Random Forest and Gradient Boosting showed a robustness in recognizing fraudulent scenarios by balancing precision and recall at the same time. All models exhibited AUC (Area Under Curve) values near 1 suggesting good classification ability according to the ROC curve which displays the proportion of true positives against the percentage of false positives. Knowing which characteristics are most useful in spotting fraudulent transactions is a crucial component of fraud detection. Using feature importance plots, we found that the most important criteria were device type, user conduct patterns, transaction frequency, transaction value and geographic location. The most comprehensible feature rankings were produced by the Random Forest and Gradient Boosting models emphasizing that fraudulent transactions frequently entail surprisingly large sums, quickly consecutive transactions or transactions from odd places. Financial institutions can use these insights to inform the development of additional behavioral analysis-based fraud prevention strategies.

Fig.6 User Interface for Detections

Fig.7 Real Time UPI Fraud Detection

Fig.8 Real Time UPI Fraud Detection

The Streamlit application's backend dynamically processes user input while loading the pre-trained detection of fraud models. The system applies the chosen fraud detection model, processes the input in real-time and returns the prediction once a user inputs transaction data. The system notifies users in the event that a transaction is identified as fraudulent enabling them to take prompt action. A probability score is another feature of the interface that lets users know how certain the model is of its prediction. Better decision-making is made possible by this feature which also increases transparency Because of their excellent accuracy, fair precision-recall trade-off and resilience to outliers Random Forest and Gradient Boosting were determined to be the top performers after all the models were evaluated. To give users real-time fraud detection capabilities, these models were stored and implemented in the Streamlit UI. In order to update the system for detecting fraud on a regular basis when new fraud patterns appear model retraining procedures were also put in place.

CONCLUSION

Using a machine learning-based UPI fraud detection system has shown to be quite successful in accurately detecting fraudulent transactions. The system minimizes false positives while accurately classifying fraudulent activity by utilizing sophisticated models including Decision Tree, Random Forest, Gradient Boosting and AdaBoost. Real-time fraud detection integrated with a Streamlit user interface improves accessibility by enabling users to evaluate transaction risks immediately. In addition to showing how well machine learning works to prevent fraud this study offers insightful information about fraud trends that helps financial institutions bolster their security protocols. But since scammers are always changing their strategies, fraud detection systems must also advance. In order to ensure a more resilient and proactive security solution future work will concentrate on integrating methods for identifying anomalies using unsupervised learning integrating deep learning models like LSTMs and Transformers to analyze sequential transaction patterns, and creating an adaptive fraud identification framework that dynamically updates with emerging fraud trends.

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